

Virtual filesystem terminal

This module is your first real practice: moving around a directory tree and listing what is there

The shell has a current directory

- A terminal session always has a current working directory
- Print it with `pwd`, print working directory
- Change it with `cd`, change directory
- List what is here with `ls`, list
- Relative paths are resolved against where you are now

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Virtual filesystem terminal

Practice navigation and file ops in a sandboxed tree. Lab mode: subset of bash — no real disk writes.

Challenge 1: Find the RTL

Navigate into the project src directory and list its files.

Not checked

Check

Next challenge

Reset filesystem

lab-shell — /home/student

Unix lab shell. Type help · Arrow keys for history.

student:~\$ try: ls · cd project · help

help

man ls

ls -la

cd rtl

less notes.txt

ln -s project/tb tb-link

tree

Tree

/

[dir] home

[dir] student <- cwd

[file] .bashrc

[file] notes.txt

[dir] project

[file] README.md

[dir] src

[file] main.v

[dir] tb

[file] test_main.v

[link] rtl -> project/src

[dir] tmp

Hints

Virtual filesystem terminal starter

Real shell practice

```
wsl - module01/examples/navigation

# Track A - real Unix (captured session)
# real shell session (Track A)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module01-vfs-terminal/examples/navigation

$ ls -la
total 0
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20 00:49 ..
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 sample_repo

$ cd sample_repo

$ ls
README.txt
build
src
tb

$ cd ..

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module01-vfs-terminal/examples/navigation
```

Real WSL shell — navigation example

Real shell practice — try these

```
# pwd — print working directory (where am I?)  
pwd
```

```
# ls -la — list all entries, long format (what is here?)  
ls -la
```

```
# cd sample_repo — change into this directory  
cd sample_repo
```

```
# ls — list names only  
ls
```

```
# cd .. — go up one directory  
cd ..
```

Pitfalls to watch

- Do not confuse a directory name with a file name when you change directories
- Remember that a leading slash starts from the filesystem root
- Hidden files start with a dot and may not show unless you ask for them with a flag like
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell
- When you are ready